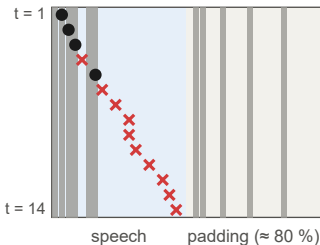


(a) One-shot retention

H2O, SnapKV, PyramidKV, PadSink-KV

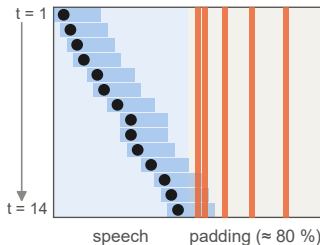
encoder positions →



(b) PadSink-Track

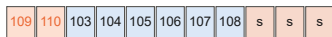
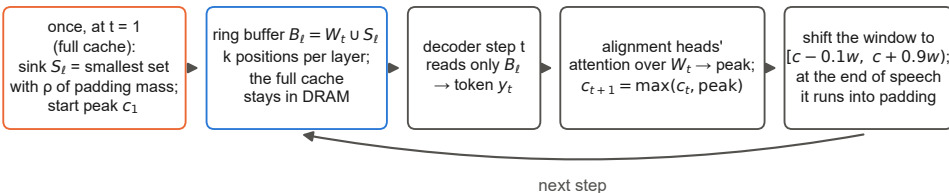
the selection is renewed at every step

encoder positions →



- position the decoder needs at step t
- ✗ needed but evicted → errors, loops
- one-shot set: chosen at $t = 1$, fixed
- window W_t : follows the alignment peak
- padding sink S_t : exact K/V, fixed

(c) One decoding step of PadSink-Track



ring buffer of one layer: slots of positions that left the window (101, 102) are overwritten with the entering ones (109, 110); in real decoding ~ 11 slots per step on average ($\sim 5\%$ of the window), with no per-step gather. s = sink slots. Attention does not depend on slot order.